Stochastic control with signatures via Riccati equations on the tensor algebra

Eduardo Abi Jaber*[1], Elie Attal†[1], and Dimitri Sotnikov‡[1, 2]

[1]Ecole Polytechnique, CMAP
[2]Engie Global Markets

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Abstract

We solve a class of non-Markovian stochastic optimal control problems with path-dependent terminal costs in semi-explicit form using path signatures. Using the Boué–Dupuis representation, we reformulate the control problem as the computation of Laplace transforms of signature functionals. Exploiting recent signature representations of such transforms on tensor algebras, we characterize the value function and optimal control through an infinite-dimensional system of Riccati equations on the extended tensor algebra. This yields an explicit feedback representation of the optimal control as a linear combination of time-extended signature coordinates with time-dependent coefficients, and enables us to establish existence and convergence of the associated signature expansions. We illustrate the approach on genuinely path-dependent and non-linear examples beyond the tractable linear-quadratic setting, including signature tracking and signature lifts of Volterra control problems.

Keywords: Path signatures, stochastic control, Riccati equation, tensor algebra.

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1 Introduction

Consider the controlled Brownian motion

$$dX_t^\alpha = \alpha_t\, dt + dW_t, \tag{1.1}$$

and the path-dependent stochastic control problem

$$\inf_\alpha \mathbb{E}\Big[\int_0^T \alpha_s^2\, ds + F\big((X_s^\alpha)_{s\le T}\big)\Big], \tag{1.2}$$

where F is a functional of the entire trajectory.

The problem is well understood with explicit solutions for certain special choices of F. For instance,

$$F((W_s)_{s\le T}) = W_T^2, \qquad \text{or} \qquad F((W_s)_{s\le T}) = \int_0^T W_s^2\, ds,$$

lead to classical linear–quadratic (LQ) control problems. In these Markovian settings, the value function is quadratic in the state variable, and the optimal control admits a linear feedback form with explicit time-dependent coefficients, which themselves solve a finite-dimensional Riccati equation. Dynamic programming yields a Hamilton–Jacobi–Bellman equation which, after a Cole–Hopf transformation, becomes linear and leads to the associated Riccati equation; see (Yong and Zhou, 1999).

*eduardo.abi-jaber@polytechnique.edu. EAJ is grateful for the financial support from the Chaires FiME-FDD and Financial Risks at Ecole Polytechnique.

†elie.attal@polytechnique.edu.

‡dmitrii.sotnikov@polytechnique.edu. DS is grateful for the financial support provided by Engie Global Markets.

A similar tractability persists in certain non-Markovian situations. For example, if

$$F((W_s)_{s\le T}) = \Big(\int_0^T K(T-s)\,dW_s\Big)^2,$$

for a locally square-integrable kernel K, one obtains a linear–quadratic Volterra control problem. Such problems can be analyzed either through variational arguments leading to linear stochastic Fredholm equations that can be solved explicitly (Abi Jaber, Neuman, and Voß, 2023), or by introducing suitable infinite-dimensional state variables that restore the Markov property and yield Riccati equations in an extended state space (Abi Jaber, Miller, and Pham, 2021).

A probabilistic viewpoint that unifies these examples is provided by the Boué and Dupuis (1998) variational representation:

$$\inf_{\alpha} \mathbb{E}\Big[\int_0^T \alpha_s^2\,ds + F((X_s^\alpha)_{s\le T})\Big] = -\log \mathbb{E}\Big[e^{-F((W_s)_{s\le T})}\Big]. \tag{1.3}$$

This identity reveals the source of tractability in the previous examples. Indeed, both the Markovian and Volterra LQ cases reduce to computing Laplace transforms of quadratic Gaussian functionals. Such quantities are explicit since they correspond to generalized chi-squared distributions and can be characterized through Fredholm (1903) determinants and resolvents.

The situation becomes much less clear for general path-dependent functionals F. While the Boué–Dupuis representation (1.3) still provides an exact characterization of the value of the control problem, allowing for its approximation via Monte Carlo methods through the right-hand side, it does not by itself provide a tractable characterization of optimal controls. Under suitable regularity assumptions on F, an optimal control can be identified with the Föllmer (1985) drift associated with the Gibbs measure

$$d\mathbb{Q}^F = \frac{e^{F(W)}}{\mathbb{E}[e^{F(W)}]}d\mathbb{P},$$

namely the unique adapted process α^* such that $W_t - \int_0^t \alpha_s^* ds$ is a Brownian motion under $\mathbb{Q}^F$. When F is Malliavin differentiable, this drift admits the explicit Clark–Ocone representation

$$\alpha_t^* = \frac{\mathbb{E}\left[e^{F(W)} D_t F(W) \mid \mathcal{F}_t\right]}{\mathbb{E}\left[e^{F(W)} \mid \mathcal{F}_t\right]}, \tag{1.4}$$

see Lehec (2013). Although explicit, this representation remains generally lacks exploitable structure for analysis or numerical computation.

In such non-Markovian settings, the main challenge is therefore to identify suitable state variables that restore tractability while preserving the path-dependent nature of the objective. Ideally, one seeks a representation that linearizes the dependence on the path, in much the same way that finite-dimensional state augmentation restores Markovianity in classical control problems.

Our contribution. The present work develops a tractable framework to tackle a broad class of non-Markovian functionals F based on path signatures. Signatures, which are sequences of iterated integrals introduced by Chen (1957), play the role of polynomials on path space and form a universal family for continuous paths. By Stone–Weierstrass approximation results, a large class of path functionals can be approximated by linear functionals of the signature. Motivated by this universality, we focus on the case where the terminal cost functional F in (1.2) is given by a finite linear combination of time-extended signatures of the controlled process X^α. Combining this representation with the conditional Boué–Dupuis formula transforms the original control problem into the computation of conditional Laplace transforms of linear signature functionals. Exploiting the algebraic structure of signatures together with recent signature expansions for log-Laplace transforms on tensor algebras derived by Abi Jaber, Attal, and Sotnikov, we show that the resulting control problem can be solved through an infinite-dimensional system of Riccati equations on the extended tensor algebra.

More precisely, our main Theorem 4.3 shows that the optimal control admits a feedback representation as a local signature expansion, whose coefficients solve a Riccati equation on the extended tensor algebra; see

(4.2). In particular, the optimal control is linear in the signature components of the controlled process, revealing a generalized affine structure reminiscent of the classical LQ setting. Moreover, when the functional reduces to a Markovian Linear–Quadratic form, our representation recovers the classical LQ formulas; see Example 4.6. More generally, in the Markovian setting beyond the LQ case, our formulas correspond precisely to the analytic expansion in the x-variable of the solution to the associated Hamilton–Jacobi–Bellman equation; see Remark 4.5. We emphasize, however, that the functional considered here is neither Linear–Quadratic nor Markovian in the controlled variable. Unlike in the classical LQ framework, our representation is intrinsically local. Recovering a global representation requires recentering the expansion in time, which leads to randomized Riccati equations with path-dependent terminal conditions; see (4.3).

We illustrate the flexibility of our framework in Section 6 through genuinely path-dependent and nonlinear examples beyond the Linear–Quadratic case, including signature tracking problems, and show that it naturally encompasses signature lifts of a broad class of Volterra control problems.

Our results provide a theoretical framework for stochastic control in non-Markovian settings beyond the Linear–Quadratic paradigm. They yield an explicit feedback representation of the optimal control in terms of the time-extended signature of the controlled process, thereby establishing a direct connection between non-Markovian stochastic control and Riccati equations on tensor algebras. In contrast to existing signature approaches, which parametrize controls via signatures and optimize over restricted classes, we stress that our method solves the original stochastic control problem over the full class of admissible progressively measurable controls. We also expect the underlying signature/Riccati structure to extend to more general control problems and dynamics, although the corresponding questions of existence and convergence for the Riccati system remain highly nontrivial, see for instance Abi Jaber, Hainaut, and Motte (2025).

Related literature. Bien relire, s'assurer que j'ai bien cité / résumé | N'hésitez pas à modifier et compléter !!

Our work connects to the growing literature on the use of signatures in stochastic control. Broadly speaking, existing approaches can be divided into three main directions.

The first direction uses signatures as feature maps, or nonlinear transformations thereof, to encode path dependence, as developed for instance in Kidger, Bonnier, Perez Arribas, Salvi, and Lyons (2019). In this approach, signatures provide finite-dimensional summaries of historical trajectories, which are then used as inputs to learning architectures or approximation schemes. For instance, Bayer, Pelizzari, and Zhu (2024) combines deep-signature and signature-kernel methods to solve non-Markovian optimal stopping problems, while Abi Jaber and Gérard (2025) uses signatures as inputs to shallow feedforward neural networks that parametrizes the optimal controls for hedging under non-Markovian stochastic volatility, showing that they can outperform recurrent architectures at a lower training cost.

The second uses signatures to parameterize admissible controls, typically through linear functionals of the signature, with coefficients chosen as optimization variables. This approach is motivated by universal approximation results, relying on Stone–Weierstrass arguments. Early examples of this viewpoint appear in Lyons, Nejad, and Perez Arribas (2020) in the context of pricing and hedging. More recently, Bank, Bayer, Hager, Riedel, and Nauen (2025) introduced general classes of signature-based controls, proved their density in the set of admissible progressively measurable controls, and proposed associated numerical methods for computing the optimal control. Financial applications and variations of this approach appear in Cuchiero and Möller (2025); Futter, Horvath, and Wiese (2023); Gennaro, Mastrolia, and Primavera (2026); Kalsi, Lyons, and Arribas (2020).

A third distinct direction, initiated by Abi Jaber, Hainaut, and Motte (2025), lifts the control problem onto signature state variables and establishes for the first time a connection between stochastic optimal control and Riccati equations on the extended tensor algebra. Under the assumption of existence of solutions to the associated Riccati system and convergence of the corresponding signature expansions, a verification theorem is derived, yielding the optimal control in feedback form as a possibly infinite linear combination of the time-extended signatures of the controlled variables, with time-dependent coefficients characterized by an infinite-dimensional Riccati equation. This framework was validated numerically in non-Markovian settings, illustrating its effectiveness in path-dependent control problems. However, the fundamental questions of existence of solutions to the Riccati system and convergence of the associated series were left open.

Our work completes this program and fully derives an optimal control in feedback form in terms of the

DS: Does "feedback control" require the uniqueness in some

signature of the controlled process. In contrast with signature-parametrization approaches, such as Lyons, Nejad, and Perez Arribas (2020); Bank, Bayer, Hager, Riedel, and Nauen (2025), which optimize over a prescribed subclass of controls with time-independent coefficients, our method solves the original stochastic control problem over the full class of admissible progressively measurable controls. The resulting optimal control is represented through genuinely time-dependent coefficients, yielding a richer class of controls and avoiding the analyticity constraints (in the time variable) implicit in time-independent parametrizations.

More broadly, related infinite-dimensional Riccati systems have also appeared in uncontrolled problems in mathematical finance, notably for computing characteristic functions of signature volatility models and signature SDEs; see Abi Jaber and Gérard (2025); Abi Jaber, Li, and Lin (2024b); Cuchiero, Svaluto-Ferro, and Teichmann (2023). A general existence and uniqueness theory for such systems, together with convergence of the associated power expansions, is delicate and has only recently been established for time-augmented Brownian signatures by Abi Jaber, Attal, and Sotnikov. Earlier partial results on the Riccati side can be found in Abi Jaber, Li, and Lin (2024b); Cuchiero and Möller (2025), while results on infinite signature expansions for Markovian and non-Markovian systems appear in Abi Jaber, Gérard, and Huang (2024a); Ben Arous (1989).

Outline. TBC

2 Preliminaries on signatures

Time-augmented signatures and extended tensor algebra. Let $(X_t)_{t\geq 0}$ be a continuous, real-valued semimartingale. We define its *time-augmented signature* $\widehat{\mathbb{X}}_{s,t}$ over $[s,t]$ as the path signature of the time-augmented process $\hat{X}_t := (t, X_t)$, that is, the collection of all iterated Stratonovich integrals

$$\widehat{\mathbb{X}}_{s,t}^{\mathbf{i_1}\dots\mathbf{i_n}} := \int_{s\leq t_1\leq\dots\leq t_n\leq t} \circ d\hat{X}_{t_1}^{i_1}\dots\circ d\hat{X}_{t_n}^{i_n},$$

for $n\geq 1$ and $i_1,\dots,i_n\in\{0,1\}$. These integrals are indexed by words $\mathbf{v}$ over the alphabet $A := \{\mathbf{0},\mathbf{1}\}$, where $\mathbf{0}$ and $\mathbf{1}$ correspond to integration against dt and $\circ dX_t$, respectively. We denote by V the set of all finite words over A, including the empty word $\varnothing$, and set $\widehat{\mathbb{X}}_{s,t}^{\varnothing} := 1$ by convention.

The time-augmented signature extends the role of monomials from classical analysis to path space: in particular, $\widehat{\mathbb{X}}_{s,t}^{\mathbf{1}\dots\mathbf{1}} = (X_t - X_s)^n/n!$ when the letter $\mathbf{1}$ appears n times. Accordingly, the signature inherits an analogous universality: any sufficiently regular functional $F((X_r - X_s)_{r\in[s,t]})$ can be approximated by a linear combination of time-augmented signature components.

Identifying the letters $\mathbf{0}$ and $\mathbf{1}$ with the standard basis vectors $(1,0)$ and $(0,1)$ of $\mathbb{R}^2$, the time-augmented signature naturally lives in the *extended tensor algebra*

$$T((\mathbb{R}^2)) := \prod_{n=0}^{+\infty}(\mathbb{R}^2)^{\otimes n} = \left\{\boldsymbol{p} = (\boldsymbol{p}_0,\boldsymbol{p}_1,\dots,\boldsymbol{p}_n,\dots) : \boldsymbol{p}_n\in(\mathbb{R}^2)^{\otimes n},\ n\geq 0\right\},$$

where $(\mathbb{R}^2)^{\otimes 0} = \mathbb{R}$ corresponds to the empty-word component. For each $n\geq 0$, the elementary tensors $\mathbf{i_1}\otimes\dots\otimes\mathbf{i_n}$, written as concatenated words $\mathbf{i_1}\dots\mathbf{i_n}$, form a natural basis of $(\mathbb{R}^2)^{\otimes n}$. Any element $\boldsymbol{p}\in T((\mathbb{R}^2))$ can thus be formally expanded as $\boldsymbol{p} = \sum_{\mathbf{v}\in V}\boldsymbol{p}^{\mathbf{v}}\,\mathbf{v}$, where the coefficients $\boldsymbol{p}^{\mathbf{v}}$ are real-valued.

We denote by $|\mathbf{v}|_{\mathbf{i}}$ the number of occurrences of the letter $\mathbf{i}\in\{\mathbf{0},\mathbf{1}\}$ in the word $\mathbf{v}$, and by $|\mathbf{v}| := |\mathbf{v}|_{\mathbf{0}} + |\mathbf{v}|_{\mathbf{1}}$ its length. For an element $\boldsymbol{p}\in T((\mathbb{R}^2))$, we define its *degree* and *partial degrees* by $\deg(\boldsymbol{p}) := \max\{|\mathbf{v}| : \boldsymbol{p}^{\mathbf{v}}\neq 0\}$ and $\deg_{\mathbf{i}}(\boldsymbol{p}) := \max\{|\mathbf{v}|_{\mathbf{i}} : \boldsymbol{p}^{\mathbf{v}}\neq 0\}$, with the convention $\deg(0) = \deg_{\mathbf{i}}(0) = -\infty$.

The extended tensor algebra is an algebra under the *tensor product*, defined component-wise by

$$(\boldsymbol{p}\otimes\boldsymbol{\ell})_n := \sum_{k=0}^{n}\boldsymbol{p}_k\otimes\boldsymbol{\ell}_{n-k},\quad n\geq 0.$$

The iterated-integral structure of the signature is naturally compatible with this product through *Chen's identity*:

$$\widehat{\mathbb{X}}_{s,u}\otimes\widehat{\mathbb{X}}_{u,t} = \widehat{\mathbb{X}}_{s,t},\quad s\leq u\leq t.$$

Linear functionals of the signature and shuffle product. Another key property of path signatures is their ability to linearize products of linear functionals, as a consequence of integration by parts. We introduce the pairing

$$\langle \boldsymbol{p}, \mathbb{X} \rangle := \sum_{\mathbf{v} \in V} \boldsymbol{p}^{\mathbf{v}} \, \mathbb{X}^{\mathbf{v}} \,, \quad \boldsymbol{p} \,, \, \mathbb{X} \in T((\mathbb{R}^2)) \,,$$

which is well-defined whenever $\boldsymbol{p}$ or $\mathbb{X}$ has finite degree, or more generally when the seminorm

$$\|\boldsymbol{p}\|_{\mathbb{X}} := \sum_{n \geq 0} \Big| \sum_{|\mathbf{v}|=n} \boldsymbol{p}^{\mathbf{v}} \, \mathbb{X}^{\mathbf{v}} \Big| \tag{2.1}$$

is finite. The time-augmented signature $\widehat{\mathbb{X}}_{s,t}$ almost surely belongs to the set of *group-like elements*:

$$G := \big\{ \mathbb{X} \in T((\mathbb{R}^2)) \, : \, \mathbb{X}^{\varnothing} = 1 \;\text{ and }\; \langle \mathbf{v}, \mathbb{X} \rangle \, \langle \mathbf{w}, \mathbb{X} \rangle = \langle \mathbf{v} \sqcup\!\sqcup \mathbf{w}, \mathbb{X} \rangle \,, \;\; \mathbf{v}, \mathbf{w} \in V \big\} \,,$$

where $\mathbf{v} \sqcup\!\sqcup \mathbf{w} \in T((\mathbb{R}^2))$ denotes the *shuffle product* of $\mathbf{v}$ and $\mathbf{w}$, a generalization of the Cauchy product defined as follows.

Definition 2.1. *The shuffle product is defined recursively by*

$$\mathbf{vi} \sqcup\!\sqcup \mathbf{wj} := (\mathbf{v} \sqcup\!\sqcup \mathbf{wj}) \otimes \mathbf{i} + (\mathbf{vi} \sqcup\!\sqcup \mathbf{w}) \otimes \mathbf{j} \,,$$

for words $\mathbf{v}, \mathbf{w} \in V$ *and letters* $\mathbf{i}, \mathbf{j} \in \{\mathbf{0}, \mathbf{1}\}$*, with the initialization* $\mathbf{v} \sqcup\!\sqcup \varnothing = \varnothing \sqcup\!\sqcup \mathbf{v} = \mathbf{v}$*.*

Concretely, $\mathbf{v} \sqcup\!\sqcup \mathbf{w}$ is the sum of all interleavings of $\mathbf{v}$ and $\mathbf{w}$ that preserve the internal ordering of each word. For instance, $\mathbf{10} \sqcup\!\sqcup \mathbf{1} = \mathbf{101} + \mathbf{110} + \mathbf{110}$.

Extending the shuffle product bilinearly to $T((\mathbb{R}^2))$ via

$$\boldsymbol{p} \sqcup\!\sqcup \boldsymbol{\ell} := \sum_{\mathbf{v}, \mathbf{w} \in V} \boldsymbol{p}^{\mathbf{v}} \, \boldsymbol{\ell}^{\mathbf{w}} \, (\mathbf{v} \sqcup\!\sqcup \mathbf{w}) \,,$$

the group-like property yields $\langle \boldsymbol{p}, \mathbb{X} \rangle \, \langle \boldsymbol{\ell}, \mathbb{X} \rangle = \langle \boldsymbol{p} \sqcup\!\sqcup \boldsymbol{\ell}, \mathbb{X} \rangle$ for $\mathbb{X} \in G$, whenever $\|\boldsymbol{p}\|_{\mathbb{X}}$ and $\|\boldsymbol{\ell}\|_{\mathbb{X}}$ are finite. This follows from the seminorm (2.1) being shuffle-compatible: $\|\boldsymbol{p} \sqcup\!\sqcup \boldsymbol{\ell}\|_{\mathbb{X}} \leq \|\boldsymbol{p}\|_{\mathbb{X}} \, \|\boldsymbol{\ell}\|_{\mathbb{X}}$.

By iterating the shuffle product, we define the *shuffle powers* $\boldsymbol{p}^{\sqcup\!\sqcup 0} := \varnothing$ and $\boldsymbol{p}^{\sqcup\!\sqcup n} := \boldsymbol{p}^{\sqcup\!\sqcup n-1} \sqcup\!\sqcup \boldsymbol{p}$, as well as the *shuffle exponential* and *shuffle logarithm*. For $\boldsymbol{p} \in T((\mathbb{R}^2))$ with $\boldsymbol{p}^{\varnothing} = 0$, they are given by

$$\exp^{\sqcup\!\sqcup}(\boldsymbol{p}) := \sum_{n \geq 0} \frac{1}{n!} \, \boldsymbol{p}^{\sqcup\!\sqcup n} \,, \quad \text{and} \quad \log^{\sqcup\!\sqcup}(\varnothing + \boldsymbol{p}) := \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} \, \boldsymbol{p}^{\sqcup\!\sqcup n} \,.$$

The condition $\boldsymbol{p}^{\varnothing} = 0$ ensures that for any fixed word $\mathbf{v}$, only finitely many terms contribute to $\exp^{\sqcup\!\sqcup}(\boldsymbol{p})^{\mathbf{v}}$ and $\log^{\sqcup\!\sqcup}(\varnothing + \boldsymbol{p})^{\mathbf{v}}$. These definitions extend to general elements: for any $\boldsymbol{p} \in T((\mathbb{R}^2))$,

$$\exp^{\sqcup\!\sqcup}(\boldsymbol{p}) := \exp(\boldsymbol{p}^{\varnothing}) \, \exp^{\sqcup\!\sqcup}(\overline{\boldsymbol{p}}) \,, \quad \text{where} \quad \overline{\boldsymbol{p}} := \boldsymbol{p} - \boldsymbol{p}^{\varnothing} \, \varnothing \,,$$

and whenever $\boldsymbol{p}^{\varnothing} > 0$,

$$\log^{\sqcup\!\sqcup}(\boldsymbol{p}) := \log(\boldsymbol{p}^{\varnothing}) \, \varnothing + \log^{\sqcup\!\sqcup}\left(\varnothing + \frac{\overline{\boldsymbol{p}}}{\boldsymbol{p}^{\varnothing}} \right) .$$

For $\mathbb{X} \in G$, these shuffle operations linearize the exponential and logarithm of linear functionals. Specifically, if $\|\boldsymbol{p}\|_{\mathbb{X}} < +\infty$, then $\exp(\langle \boldsymbol{p}, \mathbb{X} \rangle) = \langle \exp^{\sqcup\!\sqcup}(\boldsymbol{p}), \mathbb{X} \rangle$. If additionally $\boldsymbol{p}^{\varnothing} > 0$ and $\|\boldsymbol{p}\|_{\mathbb{X}} < 2\,\boldsymbol{p}^{\varnothing}$, then $\log(\langle \boldsymbol{p}, \mathbb{X} \rangle) = \langle \log^{\sqcup\!\sqcup}(\boldsymbol{p}), \mathbb{X} \rangle$; this condition corresponds to the convergence of the series $\sum_{n \geq 1} \|\overline{\boldsymbol{p}}/\boldsymbol{p}^{\varnothing}\|_{\mathbb{X}}^{n}/n$ arising from the definition of the shuffle logarithm and the shuffle compatibility of the seminorm.

Convergence classes. Since linear functionals of the signature generally involve infinite sums, their convergence requires care. In what follows, we focus on evaluating such functionals at group-like elements. An element $\boldsymbol{p} \in T((\mathbb{R}^2))$ satisfying $\|\boldsymbol{p}\|_{\mathbb{X}} < +\infty$ for all $\mathbb{X} \in G$ is the natural analogue of a power series with infinite radius of convergence. For the analogue of finite radius, we introduce

$$M_{\boldsymbol{p}}(\mathbb{X}) := \max_{1 \leq |\mathbf{v}| \leq \deg(\boldsymbol{p})} |\mathbb{X}^{\mathbf{v}}|$$

as a measure of how far $\mathbb{X}$ is from $\varnothing$ relative to the words appearing in $\boldsymbol{p}$, and the corresponding ball of radius $r > 0$ in G:

$$G_{\boldsymbol{p},r} := \left\{ \mathbb{X} \in G \,:\, M_{\boldsymbol{p}}(\mathbb{X}) \leq r \right\}.$$

A functional satisfying $\|\boldsymbol{p}\|_{\mathbb{X}} < +\infty$ for all $\mathbb{X} \in G_{\boldsymbol{p},r}$ then plays the role of a power series with radius of convergence r.

Left and right shifts. For a power series $\sum_{n\geq 0} \boldsymbol{p}_n x^n/n!$, differentiation with respect to x shifts the coefficient sequence $(\boldsymbol{p}_0, \boldsymbol{p}_1, \ldots)$ into $(\boldsymbol{p}_1, \boldsymbol{p}_2, \ldots)$. The analogue on the extended tensor algebra is the *right shift*:

$$\boldsymbol{p}|_{\mathbf{v}} := \sum_{\mathbf{w}\in V} \boldsymbol{p}^{\mathbf{w}\mathbf{v}}\, \mathbf{w}\,, \quad \mathbf{v} \in V\,.$$

These operators will appear naturally as differentiation operators when applying Itô's formula to $\langle \boldsymbol{p}, \widehat{\mathbb{X}}_t \rangle$ below. The *left shift* is defined analogously by

$${}_{\mathbf{v}}|\boldsymbol{p} := \sum_{\mathbf{w}\in V} \boldsymbol{p}^{\mathbf{v}\mathbf{w}}\, \mathbf{w}\,, \quad \mathbf{v} \in V\,.$$

By linearity, both extend to shifts by elements of the extended tensor algebra:

$$\boldsymbol{p}|_{\mathbb{X}} := \sum_{\mathbf{v}\in V} \mathbb{X}^{\mathbf{v}} \cdot \boldsymbol{p}|_{\mathbf{v}} = \sum_{\mathbf{v}\in V} \langle {}_{\mathbf{v}}|\boldsymbol{p}, \mathbb{X}\rangle \cdot \mathbf{v}\,, \quad \text{and} \quad {}_{\mathbb{X}}|\boldsymbol{p} := \sum_{\mathbf{v}\in V} \mathbb{X}^{\mathbf{v}} \cdot {}_{\mathbf{v}}|\boldsymbol{p} = \sum_{\mathbf{v}\in V} \langle \boldsymbol{p}|_{\mathbf{v}}, \mathbb{X}\rangle \cdot \mathbf{v}\,,$$

which are well-defined whenever $\|\boldsymbol{p}|_{\mathbf{v}}\|_{\mathbb{X}} < +\infty$ or $\|{}_{\mathbf{v}}|\boldsymbol{p}\|_{\mathbb{X}} < +\infty$ for all $\mathbf{v} \in V$, respectively.

If $\boldsymbol{p}$ has finite degree and $\mathbb{X}, \mathbb{Y} \in T((\mathbb{R}^2))$, then

$$\langle \boldsymbol{p}, \mathbb{X} \otimes \mathbb{Y}\rangle = \langle {}_{\mathbb{X}}|\boldsymbol{p}, \mathbb{Y}\rangle = \langle \boldsymbol{p}|_{\mathbb{Y}}, \mathbb{X}\rangle\,.$$

This shows that left and right shifts are dual to left and right tensor multiplication. Moreover, it is the tensor algebra analogue of the recentering of power series:

$$\sum_{n\geq 0} \boldsymbol{p}_n \frac{(x+y)^n}{n!} = \sum_{n\geq 0} \left(\sum_{k\geq 0} \boldsymbol{p}_{k+n} \frac{x^k}{k!} \right) \frac{y^n}{n!} =: \sum_{n\geq 0} ({}_x|\boldsymbol{p})_n \frac{y^n}{n!}\,.$$

Itô formula. We conclude this section by stating Itô's formula for linear functionals of time-augmented signatures.

Theorem 2.2. *Let $(X_t)_{t\in[0,T]}$ be a continuous semimartingale with bounded variation part $(H_t)_{t\in[0,T]}$, and let $(\boldsymbol{\ell}_t)_{t\in[0,T]}$ be a family of coefficients in $T((\mathbb{R}^2))$ such that:*

1. *for all $\mathbf{v} \in V$, $t \mapsto \boldsymbol{\ell}_t^{\mathbf{v}}$ is continuously differentiable on $[0,T]$;*
2. *$\|\boldsymbol{\ell}_t\|_{\widehat{\mathbb{X}}_t}$, $\|\dot{\boldsymbol{\ell}}_t\|_{\widehat{\mathbb{X}}_t}$, $\|\boldsymbol{\ell}_t|_{\mathbf{0}}\|_{\widehat{\mathbb{X}}_t}$, $\|\boldsymbol{\ell}_t|_{\mathbf{1}}\|_{\widehat{\mathbb{X}}_t}$, and $\|\boldsymbol{\ell}_t|_{\mathbf{11}}\|_{\widehat{\mathbb{X}}_t}$ are finite for all $t \in [0,T]$, a.s., and*

$$\int_0^T \left(\|\dot{\boldsymbol{\ell}}_t\|_{\widehat{\mathbb{X}}_t} + \|\boldsymbol{\ell}_t|_{\mathbf{0}}\|_{\widehat{\mathbb{X}}_t}\right) dt + \int_0^T \|\boldsymbol{\ell}_t|_{\mathbf{1}}\|_{\widehat{\mathbb{X}}_t}\, |dH_t| + \int_0^T \left(\|\boldsymbol{\ell}_t|_{\mathbf{11}}\|_{\widehat{\mathbb{X}}_t} + \|\boldsymbol{\ell}_t|_{\mathbf{1}}\|^2_{\widehat{\mathbb{X}}_t}\right) d\langle X\rangle_t < +\infty\,, \quad \text{a.s.}$$

Then $(\langle \boldsymbol{\ell}_t, \widehat{\mathbb{X}}_t\rangle)_{t\in[0,T]}$ is an Itô process, and for all $0 \leq s \leq t \leq T$,

$$\langle \boldsymbol{\ell}_t, \widehat{\mathbb{X}}_t\rangle = \langle \boldsymbol{\ell}_s, \widehat{\mathbb{X}}_s\rangle + \int_s^t \langle \dot{\boldsymbol{\ell}}_u + \boldsymbol{\ell}_u|_{\mathbf{0}}, \widehat{\mathbb{X}}_u\rangle\, du + \frac{1}{2}\int_s^t \langle \boldsymbol{\ell}_u|_{\mathbf{11}}, \widehat{\mathbb{X}}_u\rangle\, d\langle X\rangle_u + \int_s^t \langle \boldsymbol{\ell}_u|_{\mathbf{1}}, \widehat{\mathbb{X}}_u\rangle\, dX_u\,.$$

2.1 Left and right shifts and Itô's formula

For a word $\mathbf{w} \in V$, the left and right shifts of $\boldsymbol{p} \in T((\mathbb{C}^2))$ by $\mathbf{w}$ are defined as:

$$\boldsymbol{p}|_{\mathbf{w}} := \sum_{\mathbf{v}} \boldsymbol{p}^{\mathbf{vw}} \cdot \mathbf{v}, \quad \text{and} \quad {}_{\mathbf{w}}|\boldsymbol{p} := \sum_{\mathbf{v}} \boldsymbol{p}^{\mathbf{wv}} \cdot \mathbf{v}.$$

They act as the tensor algebra analogues of differentiation: in the power series case, right-shifting by $\mathbf{1}$ sends $\langle \boldsymbol{p}, \mathbb{x} \rangle = \sum_{n \geq 0} \boldsymbol{p}^n x^n / n!$ to $\langle \boldsymbol{p}|_{\mathbf{1}}, \mathbb{x} \rangle = \sum_{n \geq 0} \boldsymbol{p}^{n+1} x^n / n!$, corresponding to differentiation with respect to x. In the full signature setting, they arise naturally when differentiating $\langle \boldsymbol{p}, \widehat{\mathbb{X}}_{0,t} \rangle$ via Itô's formula, as detailed in Theorem 2.4. More precisely shifting $\boldsymbol{p}$ by $\mathbf{0}$ and $\mathbf{1}$ in the path-dependent linear functional $F((X_s)_{s \leq t}) := \langle \boldsymbol{p}, \widehat{\mathbb{X}}_{0,t} \rangle$ would correspond to, respectively, the time and space derivatives of F as defined in Dupire (2019). More generally, if $\boldsymbol{p}|_{\mathbf{v}} \in \mathcal{A}_{\mathbb{X}}$ for all $\mathbf{v}$, we define

$$\boldsymbol{p}|_{\mathbb{X}} := \sum_{\mathbf{v}} \mathbb{X}^{\mathbf{v}} \cdot \boldsymbol{p}|_{\mathbf{v}} = \sum_{\mathbf{v}} \langle {}_{\mathbf{v}}|\boldsymbol{p}, \mathbb{X} \rangle \cdot \mathbf{v},$$

and if ${}_{\mathbf{v}}|\boldsymbol{p} \in \mathcal{A}_{\mathbb{X}}$ for all $\mathbf{v}$,

$${}_{\mathbb{X}}|\boldsymbol{p} := \sum_{\mathbf{v}} \mathbb{X}^{\mathbf{v}} \cdot {}_{\mathbf{v}}|\boldsymbol{p} = \sum_{\mathbf{v}} \langle \boldsymbol{p}|_{\mathbf{v}}, \mathbb{X} \rangle \cdot \mathbf{v}.$$

The shift operations are dual to (left and right) tensor multiplication, as shown in the following proposition.

Proposition 2.3. *1. If $\boldsymbol{p} \in T(\mathbb{C}^2)$, and $\mathbb{X}, \mathbb{Y} \in T((\mathbb{R}^2))$, then $\langle \boldsymbol{p}, \mathbb{X} \otimes \mathbb{Y} \rangle = \langle {}_{\mathbb{X}}|\boldsymbol{p}, \mathbb{Y} \rangle = \langle \boldsymbol{p}|_{\mathbb{Y}}, \mathbb{X} \rangle$.*

2. If $\boldsymbol{p}|_{\mathbf{v}} \in \mathcal{A}_{\mathbb{X}}$ for all $\mathbf{v} \in V$, then $\langle \boldsymbol{p}, \mathbb{X} \otimes \mathbf{v} \rangle = \langle {}_{\mathbb{X}}|\boldsymbol{p}, \mathbf{v} \rangle = \langle \boldsymbol{p}|_{\mathbf{v}}, \mathbb{X} \rangle$, for all $\boldsymbol{p} \in T((\mathbb{C}^2))$.

Proof. The proof is given in Abi Jaber et al.. □

We provide a version of Itô's formula, derived by Abi Jaber et al. (2024a, Theorem 3.3) and slightly generalized for our needs.

Let $X = (X^0, X^1)$ denote a continuous semimartingale and let $H = (H^0, H^1)$ denote the corresponding bounded variation part. The following class describes the time-dependent coefficients $(\boldsymbol{\ell}_t)_{t \in [0,T]}$ (possibly infinite) such that Itô's formula can be applied to the process $t \mapsto \langle \boldsymbol{\ell}_t, \mathbb{X}_t \rangle$.

We say that $(\boldsymbol{\ell}_t)_{t \in [0,T]}$ belongs to $\mathcal{I}(X)$ if the following conditions are satisfied:

1. $||\boldsymbol{\ell}_t||_{\mathbb{X}_t} < \infty$ for $t \in [0,T]$ a.s.
2. For all $\mathbf{v} \in V$, the map $t \mapsto \boldsymbol{\ell}_t^{\mathbf{v}}$ is C^1 on $[0,T]$ and
$$\int_0^T \|\dot{\boldsymbol{\ell}}_t\|_{\mathbb{X}_t} dt < \infty \quad \text{a.s.}$$
3. For all $\mathbf{i}, \mathbf{j} \in \{\mathbf{0}, \mathbf{1}\}$ such that $\langle X^i, X^j \rangle \not\equiv 0$, we have
$$\boldsymbol{\ell}_t|_{\mathbf{i}},\ \boldsymbol{\ell}_t|_{\mathbf{ji}},\ \dot{\boldsymbol{\ell}}_t \in \mathcal{A}_{\mathbb{X}_t}, \quad t \in [0,T] \quad \text{a.s.},$$
and
$$\int_0^T ||\boldsymbol{\ell}_t|_{\mathbf{i}}||_{\mathbb{X}_t} |dH_t^i| + \int_0^T ||\boldsymbol{\ell}_t|_{\mathbf{ji}}||_{\mathbb{X}_t} d\langle X^i, X^j \rangle_t + \int_0^T ||\boldsymbol{\ell}_t|_{\mathbf{i}}||_{\mathbb{X}_t}^2 d\langle X^i \rangle_t < \infty \quad \text{a.s.}$$

Theorem 2.4 (Itô's formulas)**.** *Let $(\boldsymbol{\ell}_t)_{t \in [0,T]} \in \mathcal{I}(X)$. Then, $(\langle \boldsymbol{\ell}_t, \mathbb{X}_t \rangle)_{t \in [0,T]}$ is an Itô process and, for all $0 \leq s \leq t \leq T$,*

$$\langle \boldsymbol{\ell}_t, \mathbb{X}_t \rangle = \langle \boldsymbol{\ell}_s, \mathbb{X}_s \rangle + \int_s^t \langle \dot{\boldsymbol{\ell}}_u, \mathbb{X}_u \rangle du + \sum_{\mathbf{i} \in \{\mathbf{0},\mathbf{1}\}} \int_s^t \langle \boldsymbol{\ell}_u|_{\mathbf{i}}, \mathbb{X}_u \rangle dX_u^i + \frac{1}{2} \sum_{\mathbf{i},\mathbf{j} \in \{\mathbf{0},\mathbf{1}\}} \int_s^t \langle \boldsymbol{\ell}_u|_{\mathbf{ji}}, \mathbb{X}_u \rangle d\langle X^i, X^j \rangle_u. \tag{2.2}$$

Proof. The proof follows from (Abi Jaber et al., 2024a, Theorem 3.3 and Corollary 3.4). □

Remark 2.1. Itô's formula (2.2) extends to more general settings. For instance, it holds between two stopping times σ and τ with $0 \le \sigma \le \tau \le T$ a.s., for linear functionals whose coefficients $(\boldsymbol{\ell}_t)_{t\in[0,T]}$ are $\mathcal{F}_\sigma$-measurable. The proof is essentially unchanged and mirrors that of (Abi Jaber et al., 2024a, Theorem 3.3 and Corollary 3.4). ∎

3 Reminder on Signatures and Riccati equations

3.1 Class of admissible coefficients

TODO: Recall the class $\mathcal{B}$

3.2 Existence up to a stopping time

We recall the existence result for the Riccati equation proved in (Abi Jaber et al., Theorem 4.7, Theorem 5.7).

Theorem 3.1. *Let $\boldsymbol{p} \in \mathcal{B}$, and assume that $\mathbb{E}[\exp(\langle \boldsymbol{p}, \widehat{\mathbb{W}}_{t,T}\rangle)] \neq 0$ for all $t \in [0,T]$. There exists $\boldsymbol{\psi}$ such that fro all $\mathbf{v} \in V$, the map $t \mapsto \boldsymbol{\psi}_t^{\mathbf{v}}$ is C^1 on $[0,T]$, and*

1. *For all $t \in [0,T]$ and $\mathbb{X} \in G$ such that $\|\exp^{\sqcup\!\sqcup}(\boldsymbol{\psi}_t)\|_{\mathbb{X}} < 2|\exp(\boldsymbol{\psi}_t^{\varnothing})|$, we have $\boldsymbol{\psi}_t \in \mathcal{A}_{\mathbb{X}}$ and*

$$\mathbb{E}\left[\exp\left(\langle \boldsymbol{p}, \mathbb{X} \otimes \widehat{\mathbb{W}}_{t,T}\rangle\right)\right] = \exp(\langle \boldsymbol{\psi}_t, \mathbb{X}\rangle).$$

Modify everywhere to avoid introducing $\boldsymbol{u}_t$

2. *For any $\varepsilon \in (0,1)$, there exists $r = r(\boldsymbol{p}, T, \varepsilon) > 0$, such that $\|\boldsymbol{u}_t\|_{\mathbb{X}} \le (2-\varepsilon)|\boldsymbol{u}_t^{\varnothing}|$ for all $t \in [0,T]$ and $\mathbb{X} \in G_{\boldsymbol{p},r}$, so that $\boldsymbol{\psi}_t \in \mathcal{A}_{\boldsymbol{p},r}$. Moreover,*

$$\sup_{t\in[0,T]} \|\boldsymbol{\psi}_t\|_{\mathbb{X}} \le \sup_{t\in[0,T]} |\boldsymbol{\psi}_t^{\varnothing}| + \log\left(\varepsilon^{-1}\right) < +\infty, \quad \mathbb{X} \in G_{\boldsymbol{p},r}.$$

The Fourier–Laplace transform thus reads

$$\mathbb{E}\left[\exp\left(\langle \boldsymbol{p}, \widehat{\mathbb{W}}_T\rangle\right) \mid \mathcal{F}_t\right] = \exp(\langle \boldsymbol{\psi}_t, \widehat{\mathbb{W}}_t\rangle), \tag{3.1}$$

for all $t \in [0,T]$ such that $|M_{\boldsymbol{p}}(\widehat{\mathbb{W}}_t)| \le r$. In particular, (3.1) holds for all $t \in [\![0,\tau]\!]$, where τ is a positive stopping time given by

$$\tau := \inf\{t \ge 0 \colon\ |M_{\boldsymbol{p}}(\widehat{\mathbb{W}}_t)| \ge r\}.$$

3. *The following equation is satisfied:*

$$\begin{cases} \dot{\boldsymbol{\psi}}_t + \mathscr{R}\boldsymbol{\psi}_t = 0, & t \in [0,T] \\ \boldsymbol{\psi}_T = \boldsymbol{p} \end{cases}$$

4. *For any $\mathbf{v} \neq \varnothing$, there exists a constant $C'_{\boldsymbol{p},T,\mathbf{v}} > 0$ such that for all $\mathbb{X} \in G_{\boldsymbol{p},r}$,*

$$\sup_{t\in[0,T]} \|\boldsymbol{\psi}_t|_{\mathbf{v}}\|_{\mathbb{X}} \le C'_{\boldsymbol{p},T,\mathbf{v}} \left(\left(1 + \frac{1}{\varepsilon \inf_{t\in[0,T]} |\boldsymbol{u}_t^{\varnothing}|}\right)(1 + M_{\boldsymbol{p}}(\mathbb{X})) \exp\left(C_{\boldsymbol{p},T}(1 + M_{\boldsymbol{p}}(\mathbb{X}))^{\deg_{\mathbf{1}}(\boldsymbol{p})}\right)\right)^{|\mathbf{v}|}. \tag{3.2}$$

Theorem 3.2. *Let $\boldsymbol{p} \in \mathcal{B}$. Assuming that $\mathbb{E}[\exp(\langle \boldsymbol{p}, \widehat{\mathbb{W}}_{t,T}\rangle)] \neq 0$ for all $t \in [0,T]$, and letting $\boldsymbol{\psi}$, ε, and r be as in Theorem 3.1, we have*

Remark 3.1. add "whenever $\mathbb{X}$ is replaced by $\widehat{\mathbb{W}}_t$" ?A more tractable way to ensure the convergence is to state the result for $t < \tau$, where

$$\tau := \inf\{t \colon\ ||\boldsymbol{u}_t||_{\widehat{\mathbb{W}}_t} \ge 2\boldsymbol{u}_t^{\varnothing}\}.$$

In practice, one cannot numerically compute $\boldsymbol{u}_t$ precisely enough, so that a weaker, but numerically tractable quantity to track would be

$$\tau^{\boldsymbol{\psi}} := \inf\{t \colon e^{||\boldsymbol{\psi}_t||_{\widehat{\mathbb{W}}_t}} \ge 2e^{\boldsymbol{\psi}_t^{\varnothing}}\} = \inf\{t \colon \left|\left|\overline{\boldsymbol{\psi}_t}\right|\right|_{\widehat{\mathbb{W}}_t} \ge \log 2\}.$$

∎

3.3 Shifted development (recentering)

We also have the following flow-like property for the recentered Riccati:

$$\langle \boldsymbol{\psi}_t^{\widehat{\mathbb{X}}_u}, \widehat{\mathbb{X}}_{u,t} \rangle = \langle \boldsymbol{\psi}_t^{\widehat{\mathbb{X}}_s}, \widehat{\mathbb{X}}_{s,t} \rangle$$

for all $s, u \le t$ s.t. $M_{\boldsymbol{p}}(\widehat{\mathbb{X}}_{u,t}) < r$ and $M_{\boldsymbol{p}}(\widehat{\mathbb{X}}_{s,t}) < r$. I do not know if it is worth mentioning it.

Theorem 3.3. *Let $\boldsymbol{p} \in \mathcal{B}$ and $\mathbb{X} \in G$. Assume that $\mathbb{E}[\exp(\langle \boldsymbol{p}, \mathbb{X} \otimes \widehat{\mathbb{W}}_{t,T} \rangle)] \neq 0$ for all $t \in [0, T]$. Then*

$$\mathbb{E}\left[\exp(\langle \boldsymbol{p}, \mathbb{X} \otimes \widehat{\mathbb{W}}_{t,T} \rangle)\right] = \exp(\langle \boldsymbol{\psi}_t^{\mathbb{X}}, \varnothing \rangle), \quad t \in [0, T],$$

where $\boldsymbol{\psi}_t^{\mathbb{X}}$ is a solution to the recentered Riccati equation (3.3). In particular, if $\mathbb{E}[\exp(\langle \boldsymbol{p}, \widehat{\mathbb{W}}_T \rangle) \,|\, \mathcal{F}_t] \neq 0$ a.s., for all $t \in [0, T]$, we have

$$\mathbb{E}\left[\exp\left(\langle \boldsymbol{p}, \widehat{\mathbb{W}}_T \rangle\right) \,\middle|\, \mathcal{F}_t\right] = \exp\left(\langle \boldsymbol{\psi}_t^{\widehat{\mathbb{W}}_t}, \varnothing \rangle\right), \quad t \in [0, T].$$

DS: Formulate a more general version of recentering with expansion at $X_s, s \le t$.

To solve this issue one can recenter the development as follows:

$$\mathbb{E}\left[\exp\langle \boldsymbol{p}, \mathbb{X} \otimes \widehat{\mathbb{W}}_{t,T} \rangle\right] = \mathbb{E}\left[\exp\langle {}_{\mathbb{X}}|\boldsymbol{p}, \widehat{\mathbb{W}}_{t,T} \rangle\right] = \exp(\boldsymbol{\psi}_t^{\mathbb{X}})^{\varnothing} = (\boldsymbol{u}_t^{\mathbb{X}})^{\varnothing},$$

where

$$\begin{cases} \dot{\boldsymbol{\psi}}_t^{\mathbb{X}} + \mathscr{R}\boldsymbol{\psi}_t^{\mathbb{X}} = 0 \\ \boldsymbol{\psi}_T^{\mathbb{X}} = {}_{\mathbb{X}}|\boldsymbol{p} \end{cases} \tag{3.3}$$

which is well defined for any $\mathbb{X} \in G$.

DS: I think we no longer need this lemma

Lemma 3.4. *We have*

$$\boldsymbol{u}_t^{\mathbb{X}} = {}_{\mathbb{X}}|\boldsymbol{u}_t, \quad \mathbb{X} \in G. \tag{3.4}$$

and

$$\boldsymbol{\psi}_t^{\mathbb{X}} = {}_{\mathbb{X}}|\boldsymbol{\psi}_t, \quad \mathbb{X} \in G_{\boldsymbol{p}, r_t}. \tag{3.5}$$

Proof. For all $t \in [0, T]$ and for all $\mathbb{X}, \mathbb{Y} \in G$, we have

$$\langle \boldsymbol{u}_t, \mathbb{X} \otimes \mathbb{Y} \rangle = \langle \boldsymbol{u}_t^{\mathbb{X}}, \mathbb{Y} \rangle, \tag{3.6}$$

by definition of $\boldsymbol{u}_t^{\mathbb{X}}$. We will show that

$$\langle \boldsymbol{u}_t|_{\mathbf{v}}, \mathbb{X} \otimes \mathbb{Y} \rangle = \langle \boldsymbol{u}_t^{\mathbb{X}}|_{\mathbf{v}}, \mathbb{Y} \rangle, \quad \mathbb{X}, \mathbb{Y} \in G, \tag{3.7}$$

for all $\mathbf{v} \in V$. Taking $\mathbb{Y} = \varnothing$ would imply (3.4). We prove (3.7) by induction in $|\mathbf{v}|$. For $\mathbf{v} = \varnothing$, the result coincides with (3.6). Suppose now that (3.7) holds for $\mathbf{v} \in V$ and set $\mathbb{Y}_s = \mathbb{Y} \otimes \exp^{\otimes}(s \cdot \mathbf{i})$, $s \in [0, \delta]$, $\mathbb{Y} \in G, \mathbf{i} \in A$ and write the dynamics for the both sides of (3.7):

$$\langle \boldsymbol{u}_t|_{\mathbf{v}}, \mathbb{X} \otimes \mathbb{Y} \rangle + \int_0^s \langle (\boldsymbol{u}_t|_{\mathbf{v}})|_{\mathbf{i}}, \mathbb{X} \otimes \mathbb{Y}_u \rangle du = \langle \boldsymbol{u}_t^{\mathbb{X}}|_{\mathbf{v}}, \mathbb{Y} \rangle + \int_0^s \langle (\boldsymbol{u}_t^{\mathbb{X}}|_{\mathbf{v}})|_{\mathbf{i}}, \mathbb{Y}_u \rangle du \tag{3.8}$$

By Theorem 2.4, to apply this representation, one should justify that

$$\int_0^\delta ||\boldsymbol{u}_t|_{\mathbf{v}}||_{\mathbb{X} \otimes \mathbb{Y}_u} \, du < \infty, \quad \int_0^\delta \|\boldsymbol{u}_t^{\mathbb{X}}|_{\mathbf{v}}\|_{\mathbb{Y}_u} du < \infty,$$

which is true thanks to THE ENORMOUS BOUND allowing us to obtain a uniform in u bound for the norms.

The first terms in (3.8) coincide by the induction assumption, while the equality of the second ones imply

$$\langle \boldsymbol{u}_t|_{\mathbf{iv}}, \mathbb{X} \otimes \mathbb{Y}_u \rangle = \langle \boldsymbol{u}_t^{\mathbb{X}}|_{\mathbf{iv}}, \mathbb{Y}_u \rangle,$$

for a.e. $u \in [0,\delta]$. Moreover, the applications $u \mapsto \langle \boldsymbol{u}_t|_{\mathbf{iv}}, \mathbb{X} \otimes \mathbb{Y}_u\rangle$ and $u \mapsto \langle \boldsymbol{u}_t^{\mathbb{X}}|_{\mathbf{iv}}, \mathbb{Y}_u\rangle$ are continuous as uniformly convergent functional series (always thanks to THE ENORMOUS BOUND). Hence, taking $u \to 0$, we obtain

$$\langle \boldsymbol{u}_t|_{\mathbf{iv}}, \mathbb{X} \otimes \mathbb{Y}\rangle = \langle \boldsymbol{u}_t^{\mathbb{X}}|_{\mathbf{iv}}, \mathbb{Y}\rangle.$$

Recalling that the letter $\mathbf{i}$ and the element $\mathbb{Y} \in G$ were arbitrary, we conclude the proof of (3.7) and, hence, the proof of (3.4).

The proof of the second part (3.5) is analogous, though it requires proving (3.7) for $\mathbb{X}, \mathbb{Y} \in G$ such that $M_{\boldsymbol{p}}(\mathbb{X} \otimes \mathbb{Y}) < r_t$ and taking δ small enough to ensure that $M_{\boldsymbol{p}}(\mathbb{X} \otimes \mathbb{Y}_u) < r_t$ for all $u \in [0,\delta]$. □

4 Optimal control and value function

4.1 Weak optimal control problem statement

In what follows, we set

$$\Omega = \{\omega \in C([0,T],\mathbb{R})\colon\ \omega(0) = 0\},$$

and let $X = (X_t)_{t\in[0,T]}$ be the coordinate process: $X_t(\omega) = \omega(t)$. Let $\mathbb{P}$ denote Wiener measure and $\mathbb{F} = (\mathcal{F}_t)_{t\in[0,T]}$ be the canonical filtration.

We now specify the weak control problem formulation.

Definition 4.1. *An admissible weak control is a tuple*

$$\pi = (\mathbb{Q}, W^{\mathbb{Q}}, \alpha),$$

such that $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{Q})$ is a filtered probability space satisfying the usual conditions, $W^{\mathbb{Q}}$ is a $\mathbb{Q}$-Brownian motion, and α is $\mathbb{F}$-adapted with $\mathbb{E}^{\mathbb{Q}}[\int_0^T |\alpha_t|^2\,dt] < \infty$. For an admissible weak control π, the controlled variable X^α is defined by

$$X_t^\alpha := W_t^{\mathbb{Q}} + \int_0^t \alpha_s\,ds.$$

The class of admissible controls is denoted by $\mathcal{A}$.

In what follows, we use π and α interchangeably.

Let F be a measurable functional $F\colon C([0,T],\mathbb{R}) \to \mathbb{R}$. The weak optimal control problem reads

$$\sup_{\alpha\in\mathcal{A}} J(\alpha) = \sup_{\alpha\in\mathcal{A}} \mathbb{E}^{\mathbb{Q}}\left[-\frac{1}{2}\int_0^T |\alpha_t|^2\,dt + F(X^\alpha)\right]. \tag{4.1}$$

minimiser (inf) plutôt que maximiser partout ?

Definition 4.2. *The value process corresponding to (4.1) is defined by*

$$\psi_t^* = \operatorname*{ess\,sup}_{\alpha\in\mathcal{A}_{t,T}} \mathbb{E}^{\mathbb{Q}}\left[-\frac{1}{2}\int_t^T |\alpha_s|^2\,ds + F(X^\alpha)\,\middle|\,\mathcal{F}_t\right],$$

where $\mathcal{A}_{t,T} = \dots$

DS: Il y a trop de − qui apparaissent dans ce cas, non?

4.2 Main result

Theorem 4.3. *Let $F(X^\alpha) = \langle \boldsymbol{p}, \widehat{\mathbb{X}}_T^\alpha\rangle$ for some $\boldsymbol{p} \in \mathcal{B}$. Denote*

$$\tau^{t_0} = \inf\{t > t_0\colon\ M_{\boldsymbol{p}}(\widehat{\mathbb{X}}_{t_0,t}^{\alpha^*}) \geq r\} \wedge T, \quad t_0 \in [0,T],$$

and let $\boldsymbol{\psi}^{\mathbb{X}}$ be the solution to the recentered Riccati equation (3.3). Then, for all $t_0 \in [0,T]$, the value process ψ^* and the optimal control α^*, corresponding to the weak optimal control problem (4.1), are given on $[\![t_0, \tau^{t_0}]\!]$ by

$$\psi_t^* = \langle \boldsymbol{\psi}^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}}, \widehat{\mathbb{X}}_{t_0,t}^{\alpha^*}\rangle, \qquad \alpha_t^* = \langle \boldsymbol{\psi}_t^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}}|_{\mathbf{1}}, \widehat{\mathbb{X}}_{t_0,t}^{\alpha^*}\rangle, \qquad t \in [\![t_0, \tau^s]\!]. \tag{4.2}$$

In particular, for all $t \in [0,T]$, we have

$$\psi_t^* = \langle \boldsymbol{\psi}^{\widehat{\mathbb{X}}_t^{\alpha^*}}, \varnothing \rangle, \qquad \alpha_t^* = \langle \boldsymbol{\psi}_t^{\widehat{\mathbb{X}}_t^{\alpha^*}} |_{\mathbf{1}}, \varnothing \rangle, \qquad t \in [0,T]. \tag{4.3}$$

The controlled variable $X_t^{\alpha^*} = W_t^{\mathbb{Q}^*} + \int_0^t \alpha_s^* \, ds$ is a weak solution to the following signature-SDE

$$dX_t^{\alpha^*} = \langle \boldsymbol{\psi}_t^{\widehat{\mathbb{X}}_t^{\alpha^*}} |_{\mathbf{1}}, \varnothing \rangle \, dt + dW_t^{\mathbb{Q}^*}, \quad t \in [0,T]. \tag{4.4}$$

Remark 4.1. peut etre une remarque ou on explique avec des mots, la rep local et la rep globale... le control est analytique en quelque sorte en la signature mais pas entire, rayon de conv fini en general. Si on prend $t_0 = 0$ riccati determinisite etc... ∎

Remark 4.2. Theorem 4.3 provides an explicit form of the functional Φ_t appearing in Lemma 4.6, hence showing that the optimal control is closed-loop with an explicit signature-Markovian feedback functional, computable via the Riccati equation. ∎

Corollary 4.4. *Dimitri* *Show that by writing $\boldsymbol{p} = ...$, we recover running cost + terminal cost:*

$$\inf_{\alpha} \mathbb{E}\left[\frac{1}{2}\int_0^T \alpha_t^2 \, dt + \langle \boldsymbol{p}, \widehat{\mathbb{W}}_T^{\alpha} \rangle + \int_0^T \langle \boldsymbol{q}, \widehat{\mathbb{W}}_s^{\alpha} \rangle ds\right]$$

the Riccai equation and optimakl control can be re-written with source term, (comment that is more stable numerically?)

Remark 4.3. In the signature control problem $F(X^{\alpha}) = \langle \boldsymbol{p}, \widehat{\mathbb{X}}_T^{\alpha} \rangle$, $\boldsymbol{p} \in \mathcal{B}$, the feedback control functional can be equivalently written in terms of expectations of functionals of the Brownian signature. Indeed, recalling the Clark–Ocone representation (1.4) of the optimal control, and using the fact that the Malliavin derivative of the Brownian signature is given by

$$D_t \widehat{\mathbb{W}}_T = \widehat{\mathbb{W}}_t \otimes \mathbf{1} \otimes \widehat{\mathbb{W}}_{t,T},$$

as shown in Abi Jaber et al. (2026, Corollary 4.7), we obtain that the value process and the optimal control admit the following feedback form:

$$\psi(t, \widehat{\mathbb{X}}_t) = \log \mathbb{E}^{\mathbb{Q}^*}[\exp(\langle \boldsymbol{p}, \widehat{\mathbb{X}}_t \otimes \widehat{\mathbb{W}}_{t,T}^{\mathbb{Q}^*} \rangle)], \qquad \alpha^*(t, \widehat{\mathbb{X}}_t) = \frac{\mathbb{E}^{\mathbb{Q}^*}[\exp(\langle \boldsymbol{p}, \widehat{\mathbb{X}}_t \otimes \widehat{\mathbb{W}}_{t,T}^{\mathbb{Q}^*} \rangle) \langle \boldsymbol{p}, \widehat{\mathbb{X}}_t \otimes \mathbf{1} \otimes \widehat{\mathbb{W}}_{t,T}^{\mathbb{Q}^*} \rangle]}{\mathbb{E}^{\mathbb{Q}^*}[\exp(\langle \boldsymbol{p}, \widehat{\mathbb{X}}_t \otimes \widehat{\mathbb{W}}_{t,T}^{\mathbb{Q}^*} \rangle)]}, \tag{4.5}$$

reducing the simulation to that of the Brownian signature $\widehat{\mathbb{W}}_{t,T}$ only. The disadvantage of this method is the necessity to run the Monte Carlo simulation (4.5) at each time $t \in [0,T]$, while the Riccati approach of Theorem 4.3 allows one to solve the Riccati equation once at some $t_0 \in [0,T]$ to determine the coefficients $(\boldsymbol{\psi}_t^{\widehat{\mathbb{X}}_{t_0}})_{t \in [t_0,T]}$, which can be used as long as the convergence condition $M_{\boldsymbol{p}}(\widehat{\mathbb{X}}_{t_0,t}) \leq r$ is satisfied. ∎

eaj est ce la bonne terminologie ? feedback? open loop? closed loop?

Remark 4.4. We expect our methodology to extend naturally to time-dependent coefficients, that is, by replacing the running cost $\langle \boldsymbol{q}, \widehat{\mathbb{X}}_s^{\alpha} \rangle$ in Corollary 4.4 with $\langle \boldsymbol{q}_s, \widehat{\mathbb{X}}_s^{\alpha} \rangle$, where $\boldsymbol{q}_s \in \mathcal{B}$ for each $s \in [0,T]$. In this case, we conjecture that the optimal control retains the same feedback form as in (??), with ψ solving the Riccati equation (??) but with time-dependent input $(\boldsymbol{q}_t)_{t \in [0,T]}$. Establishing this extension would require a corresponding generalization of the log-Laplace transform results of Abi Jaber, Attal, and Sotnikov to the time-dependent setting. Such an extension could be particularly useful for representing certain non-Markovian models within our framework while reducing dimensional complexity, as it would avoid the need to expand in the time variable, for instance in the case of Volterra structures, see for instance (Abi Jaber, Gérard, and Huang, 2024a, Section 4.3).

EAJ: j'ai rédigé la rmk ;)

Corollary 4.5 (Markovian case)**.** *Let $p, q : \mathbb{R} \to \mathbb{R}$ be a polynomial*

$$p(x) = \sum_{n=0}^{N} p^k \frac{x^k}{k!} \quad q(x) = \sum_{n=0}^{M} q^k \frac{x^k}{k!},$$

for some $p^k, q^k \in \mathbb{R}$, with even degrees N, M and leading coefficients $p_N, q_N < 0$. Then, the optimal control with terminal cost $p(W_T^{\alpha})$ and running cost $\int_0^T q(W_t^{\alpha}) dt$... admits a solution... value function as power sries and optimal control analytic in W_t^{α} give form. and show that Riccati simplify with Cauchy product...

Proof. .... □

Remark 4.5. Link with pde in the case of cor... our result provides an analytic expansion of the associated HJB equation. Write the Hjb

Example 4.6. [Linear-Quadratic case] Show that when $N, M \leq 2$, we recover the standard LQ setting with Riccati equations finite dimensional and give value fun and control. ∎

4.3 Proof of the main result

(eaj: ici (dans (4.8)) c'est X ou X^{α^*}? comme on melange $\mathbb{P}$ et $\mathbb{P}^*$, est que les mesures sont equivalentes ? peut etre precise $\hat{P}$-as etc si besoin...)

Oui, on a $\mathbb{P}^* \sim \mathbb{P}$ et $X = W^{\mathbb{P}^*} + \int \alpha^*$. Du coup, le processus canonique X (qui est fixé au début de la section) est la variable controlée sous $\mathbb{P}^*$. X brownien sous $\mathbb{P}$? pourquoi ne pas l'appeler W? et X deviendrait X^α?

Oui, X est un $\mathbb{P}$-BM. cad il deviendrait ? C'est une fonction $X_t \colon \Omega \to \mathbb{R}$ qui ne change pas quand on change la mesure, mais dont la loi change et devient celle de BM + drift opti.

Lemma 4.6. *Assume that* $\mathbb{E}^{\mathbb{P}}[e^{F(X)}] < \infty$ *and* $\mathbb{E}^{\mathbb{P}}[F^+(X)e^{F(X)}] < \infty$. *Then, there exists a weak control* $\pi^* = (\mathbb{Q}^*, W^{\mathbb{Q}^*}, \alpha^*)$ *with the controlled variable* $X^{\alpha^*} = X$ *which is a solution to the weak optimal control problem* (4.1), *and*

$$\sup_{\alpha \in \mathcal{A}} J(\alpha) = J(\alpha^*) = \log \mathbb{E}^{\mathbb{P}}[e^{F(X)}]. \tag{4.6}$$

The value process ψ^* *is given by* $\psi_t^* = \log \mathbb{E}^{\mathbb{P}}[e^{F(X)} \,|\, \mathcal{F}_t]$ *and admits the backward representation*

$$\psi_t^* = F(X^{\alpha^*}) - \int_t^T \alpha_s^* \, dW_s^{\mathbb{Q}^*} - \frac{1}{2} \int_t^T |\alpha_s^*|^2 \, ds, \quad t \in [0, T]. \tag{4.7}$$

Moreover, the optimal control is closed-loop (adapted to the state): there exists a non-anticipative measurable functional $\Phi_t(\cdot)$ *such that*

$$\alpha_t^* = \Phi_t(X_s^{\alpha^*}, \ s \in [0, t]). \tag{4.8}$$

Proof. We first show that $\sup_{\alpha \in \mathcal{A}} J(\alpha) \leq \log Z$, where we denoted $Z := \mathbb{E}^{\mathbb{P}}[e^{F(X)}]$. Consider an arbitrary admissible control $\pi = (\mathbb{Q}, W^{\mathbb{Q}}, \alpha) \in \mathcal{A}$ and define the measure $\mathbb{Q}'$ such that

$$\frac{d\mathbb{Q}'}{d\mathbb{Q}} = \exp\left(- \int_0^T \alpha_s \, dW_s^{\mathbb{Q}} - \frac{1}{2} \int_0^T |\alpha_s|^2 \, ds \right) =: N_T.$$

Since the Doléans–Dade exponential N on the r.h.s. is a positive local martingale, it is a supermartingale, and we have $\mathbb{E}^{\mathbb{Q}}\left[\frac{d\mathbb{Q}'}{d\mathbb{Q}}\right] \leq 1$ and $\mathbb{Q}' \ll \mathbb{Q}$ (though it is not necessarily a probability measure). Localizing N with $(\sigma_n)_{n \geq 1}$, we construct a sequence of probability measures $d\mathbb{Q}'_n = N_{T \wedge \sigma_n} \, d\mathbb{Q}$. It follows from Girsanov's theorem that

$$X_t^{\alpha,n} = W_t^{\mathbb{Q}} + \int_0^{t \wedge \sigma_n} \alpha_s \, ds \quad \text{is a } \mathbb{Q}'_n\text{-Brownian motion},$$

so that

$$Z = \mathbb{E}^{\mathbb{P}}[e^{F(X)}] = \mathbb{E}^{\mathbb{Q}'_n}[e^{F(X^{\alpha,n})}] = \mathbb{E}^{\mathbb{Q}}[N_{T \wedge \sigma_n} e^{F(X^{\alpha,n})}].$$

Since $X^{\alpha,n} = X^\alpha$ from some index $n \geq n^*(\omega)$ on, we have $N_{T \wedge \sigma_n} e^{F(X^{\alpha,n})} \to N_T e^{F(X^\alpha)}$ pointwise. By Fatou's lemma, we obtain

$$\mathbb{E}^{\mathbb{Q}}[N_T e^{F(X^\alpha)}] \leq \mathbb{E}^{\mathbb{P}}[e^{F(X)}] = Z. \tag{4.9}$$

Finally, it follows from (4.9) and Jensen's inequality that

$$\log Z \geq \log \mathbb{E}^{\mathbb{Q}}\left[\exp\left(- \int_0^T \alpha_t \, dW_t^{\mathbb{Q}} - \frac{1}{2} \int_0^T |\alpha_t|^2 \, dt + F(X^\alpha) \right) \right] \geq \mathbb{E}^{\mathbb{Q}}\left[-\frac{1}{2} \int_0^T |\alpha_t|^2 \, dt + F(X^\alpha) \right] = J(\alpha),$$

where we used that $\mathbb{E}^{\mathbb{Q}}[\int_0^T |\alpha_t|^2 \, dt] < \infty$ by the definition of admissibility. This proves $\sup_{\alpha \in \mathcal{A}} J(\alpha) \leq \log Z$.

We now construct $\pi^* \in \mathcal{A}$ for which the optimal value $\log Z$ is attained. The construction of α^* follows the proof of (Lehec, 2013, Theorem 2) verbatim until the step where we add an argument yielding the representation (4.7), starting (4.10) below. We define a measure $\mathbb{Q}^*$ associated to the Radon–Nikodym derivative

$$\frac{d\mathbb{Q}^*}{d\mathbb{P}} := \frac{e^{F(X)}}{Z}, \qquad M_t := \mathbb{E}^{\mathbb{P}}\left[\frac{d\mathbb{Q}^*}{d\mathbb{P}} \,\middle|\, \mathcal{F}_t\right].$$

Note that M is an a.s. positive $\mathbb{P}$-martingale (since $e^{F(X)} > 0$ a.s.) with $M_0 = 1$ and $M_T = \frac{e^{F(X)}}{Z}$. By the martingale representation theorem, there exists an $\mathbb{F}$-predictable process h such that $\mathbb{P}\left(\int_0^T |h_t|^2\, dt < \infty\right) = 1$, and

$$M_t = 1 + \int_0^t h_s\, dX_s = 1 + \int_0^t \alpha_s^* M_s\, dX_s, \quad t \in [0,T],$$

where we defined

$$\alpha_t^* := \frac{h_t}{M_t} \mathbb{1}_{\{M_t > 0\}}.$$

The process α^* is adapted and verifies $\mathbb{P}\left(\int_0^T |\alpha_t^*|^2\, dt < \infty\right) = 1$, see the proof of (Lehec, 2013, Theorem 2) for details. Since M is a true martingale, Girsanov's theorem applies: the process

$$W_t^{\mathbb{Q}^*} = X_t - \int_0^t \alpha_s^*\, ds \quad \text{is a } \mathbb{Q}^*\text{-Brownian motion.}$$

Hence, the optimal control candidate is $\pi^* = (\mathbb{Q}^*, W^{\mathbb{Q}^*}, \alpha^*)$ with the corresponding controlled variable $X^{\alpha^*} = W^{\mathbb{Q}^*} + \int_0^\cdot \alpha_s\, ds = X$ coinciding with the coordinate process.

We define the process $\psi_t := \log \mathbb{E}^{\mathbb{P}}[e^{F(X)} \,|\, \mathcal{F}_t]$. Applying Itô's lemma yields

$$\begin{aligned} \psi_t = \log(ZM_t) &= \log Z + \int_0^t \alpha_s^*\, dX_s - \frac{1}{2}\int_0^t |\alpha_s^*|^2\, ds \\ &= \log Z + \int_0^t \alpha_s^*\, dW_s^{\mathbb{Q}^*} + \frac{1}{2}\int_0^t |\alpha_s^*|^2\, ds, \quad t \in [0,T], \end{aligned} \tag{4.10}$$

with $\psi_T = F(X)$. Note that since M is a positive $\mathbb{P}$-martingale, M^{-1} is a $\mathbb{Q}^*$-martingale, so that $\psi_t = \log Z - \log M_t^{-1}$ is a $\mathbb{Q}^*$-submartingale. Localizing α^* with $(\tau_n)_{n\geq 1}$ and taking the expectation $\mathbb{E}^{\mathbb{Q}^*}$, we obtain

$$\mathbb{E}^{\mathbb{Q}^*}\left[\frac{1}{2}\int_0^{T\wedge\tau_n} |\alpha_s^*|^2\, ds\right] = \mathbb{E}^{\mathbb{Q}^*}\left[\psi_{T\wedge\tau_n}\right] - \log Z = \mathbb{E}^{\mathbb{Q}^*}\left[\psi^+_{T\wedge\tau_n}\right] - \mathbb{E}^{\mathbb{Q}^*}\left[\psi^-_{T\wedge\tau_n}\right] - \log Z. \tag{4.11}$$

The l.h.s. converges to $\mathbb{E}^{\mathbb{Q}^*}\left[\frac{1}{2}\int_0^T |\alpha_s^*|^2\, ds\right]$ by the monotone convergence theorem. For the negative part of $\psi_{T\wedge\tau_n}$, we have

$$\mathbb{E}^{\mathbb{Q}^*}\left[\psi^-_{T\wedge\tau_n}\right] = \mathbb{E}^{\mathbb{P}}\left[M_{T\wedge\tau_n} \log^-(ZM_{T\wedge\tau_n})\right] = \mathbb{E}^{\mathbb{P}}\left[-M_{T\wedge\tau_n} \log(ZM_{T\wedge\tau_n}) \mathbb{1}_{\{ZM_{T\wedge\tau_n} < 1\}}\right] \leq Z^{-1}e^{-1},$$

since $-x\log(x) \leq e^{-1}$ on $(0,1]$. Hence, by the dominated convergence theorem, $\mathbb{E}^{\mathbb{Q}^*}\left[\psi^-_{T\wedge\tau_n}\right] \to \mathbb{E}^{\mathbb{Q}^*}\left[\psi^-_T\right] < \infty$ as $n \to +\infty$. For the positive part, we use the submartingale property of ψ: $\psi_{T\wedge\tau_n} \leq \mathbb{E}^{\mathbb{Q}^*}\left[\psi_T \,|\, \mathcal{F}_{T\wedge\tau_n}\right]$, so that

$$\psi^+_{T\wedge\tau_n} \leq \mathbb{E}^{\mathbb{Q}^*}\left[\psi_T \,|\, \mathcal{F}_{T\wedge\tau_n}\right]^+ \leq \mathbb{E}^{\mathbb{Q}^*}\left[\psi^+_T \,|\, \mathcal{F}_{T\wedge\tau_n}\right] = \mathbb{E}^{\mathbb{Q}^*}\left[F^+(X) \,|\, \mathcal{F}_{T\wedge\tau_n}\right].$$

As $\mathbb{E}^{\mathbb{Q}^*}[P^+(X)] = \mathbb{E}^{\mathbb{P}}[P^+(X)e^{F(X)}] < \infty$ by assumption, the r.h.s. is $\mathbb{Q}^*$-uniformly integrable and hence $(\psi^+_{T\wedge\tau_n})_{n\geq 1}$ is $\mathbb{Q}^*$-uniformly integrable as well. This implies $\mathbb{E}^{\mathbb{Q}^*}\left[\psi^+_{T\wedge\tau_n}\right] \to \mathbb{E}^{\mathbb{Q}^*}\left[\psi^+_T\right] < \infty$ as $n \to +\infty$. Passing to the limit in (4.11), we conclude

$$\mathbb{E}^{\mathbb{Q}^*}\left[\frac{1}{2}\int_0^T |\alpha_s^*|^2\, ds\right] = \mathbb{E}^{\mathbb{Q}^*}\left[\psi_T\right] - \log Z = \mathbb{E}^{\mathbb{Q}^*}\left[F(X)\right] - \log Z < \infty. \tag{4.12}$$

In particular, this implies

$$J(\alpha^*) = \mathbb{E}^{\mathbb{Q}^*}\left[-\frac{1}{2}\int_0^T |\alpha_s^*|^2\, ds + F(X)\right] = \log Z,$$

eaj: il faut retirer l'indicatric dans *alpha** du coup

DS: Mais on peut avoir les ω de mésure 0 où la quantité n'est pas déf. C'est pas grave?

which yields the second equality in (4.6).

We now identify the value process. Repeating the argument above with conditional expectations in place of expectations and the conditional Girsanov transform in place of Girsanov's theorem, one obtains

$$\mathbb{E}^{\mathbb{Q}}\left[-\frac{1}{2}\int_t^T |\alpha_s|^2\,ds + F(X^\alpha)\,\middle|\,\mathcal{F}_t\right] \le \log \mathbb{E}^{\mathbb{P}}\left[e^{F(X)}\,\middle|\,\mathcal{F}_t\right], \quad a.s.,$$

for all $\pi = (\mathbb{Q}, W_s^{\mathbb{Q}}, \alpha) \in \mathcal{A}_{t,T}$, and

$$\mathbb{E}^{\mathbb{Q}^*}\left[-\frac{1}{2}\int_t^T |\alpha_s^*|^2\,ds + F(X)\,\middle|\,\mathcal{F}_t\right] = \log \mathbb{E}^{\mathbb{P}}\left[e^{F(X)}\,\middle|\,\mathcal{F}_t\right], \quad a.s.$$

By the definition of the essential supremum, $\psi_t^* = \log \mathbb{E}^{\mathbb{P}}\left[e^{F(X)}\,\middle|\,\mathcal{F}_t\right] = \psi_t$. The of the backward equation stated in the theorem follows from (4.10).

Finally, (4.8) follows from the fact that α^* is $\mathbb{F}$-predictable by construction and that $\mathbb{F}$ is generated by the canonical process $X = X^{\alpha^*}$. □

Remark 4.7. The measure $\mathbb{Q}^*$ constructed in the proof of Theorem 4.6 is the unique maximizer in the relative entropy duality

$$\sup_{\mathbb{Q}\ll\mathbb{P}} \left\{\mathbb{E}^{\mathbb{Q}}[F(X)] - H(\mathbb{Q}\,|\,\mathbb{P})\right\} = \log \mathbb{E}^{\mathbb{P}}[e^{F(X)}]. \tag{4.13}$$

Indeed, by construction, $\mathbb{Q}^* \sim \mathbb{P}$, and the relative entropy

$$H(\mathbb{Q}^*\,|\,\mathbb{P}) = \mathbb{E}^{\mathbb{Q}^*}\left[\log\left(\frac{d\mathbb{Q}^*}{d\mathbb{P}}\right)\right] = \mathbb{E}^{\mathbb{Q}^*}\left[\psi_T\right] - \log \mathbb{E}^{\mathbb{P}}[e^{F(X)}] = \mathbb{E}^{\mathbb{Q}^*}\left[\frac{1}{2}\int_0^T |\alpha_s^*|^2\,ds\right] < \infty,$$

by (4.12), so that

$$\mathbb{E}^{\mathbb{Q}^*}[F(X)] - H(\mathbb{Q}^*\,|\,\mathbb{P}) = \mathbb{E}^{\mathbb{Q}^*}\left[-\frac{1}{2}\int_0^T |\alpha_s^*|^2\,ds + F(X)\right] = \log \mathbb{E}^{\mathbb{P}}[e^{F(X)}],$$

by (4.6). The uniqueness of the maximizer follows from the strict concavity of the objective functional in (4.13). ∎

Proof of Theorem 4.3. We recall that $X^{\alpha^*} = X$ is a $\mathbb{P}$-Brownian motion. By Lemma 4.6, the value process is given by the log-conditional Laplace transform, which, by Theorem 3.3, can be written for some $t_0 \in [0,T]$ as

$$\psi_t^* = \log \mathbb{E}^{\mathbb{P}}[e^{\langle \boldsymbol{p}, \widehat{\mathbb{X}}_T^{\alpha^*}\rangle}\,|\,\mathcal{F}_t] = \langle \boldsymbol{\psi}_t^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}}, \widehat{\mathbb{X}}_{t_0,t}^{\alpha^*}\rangle, \tag{4.14}$$

whenever $M_{\boldsymbol{p}}(\widehat{\mathbb{X}}_{t_0,t}^{\alpha^*}) \le r$, or, equivalently, whenever $t \in [\![t_0, \tau^{t_0}]\!]$.

The bounds (3.2) allows one to apply Itô's formula (given by Theorem 2.4) to $t \mapsto \langle \boldsymbol{\psi}_t^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}}, \widehat{\mathbb{X}}_{t_0,t}^{\alpha^*}\rangle$ between t_0 and $t \wedge \tau^{t_0}$:

$$\psi_{t\wedge\tau^{t_0}} = \psi_{t_0} + \int_{t_0}^{t\wedge\tau^{t_0}} \langle \dot{\boldsymbol{\psi}}_s^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}} + \mathscr{L}\boldsymbol{\psi}_s^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}}, \widehat{\mathbb{X}}_{t_0,s}^{\alpha^*}\rangle\,ds + \int_{t_0}^{t\wedge\tau^{t_0}} \langle \boldsymbol{\psi}_s^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}}|_{\mathbf{1}}, \widehat{\mathbb{X}}_{t_0,s}^{\alpha^*}\rangle\,dX_s.$$

Applying the Riccati equation (3.3) yields

$$\begin{aligned}\psi_{t\wedge\tau^s} &= \psi_{t_0} - \frac{1}{2}\int_{t_0}^{t\wedge\tau^{t_0}} \langle \boldsymbol{\psi}_s^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}}|_{\mathbf{1}}, \widehat{\mathbb{X}}_{t_0,s}^{\alpha^*}\rangle^2\,ds + \int_{t_0}^{t\wedge\tau^{t_0}} \langle \boldsymbol{\psi}_s^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}}|_{\mathbf{1}}, \widehat{\mathbb{X}}_{t_0,s}^{\alpha^*}\rangle\,dX_s \\ &= \psi_{t_0} + \frac{1}{2}\int_{t_0}^{t\wedge\tau^{t_0}} \langle \boldsymbol{\psi}_s^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}}|_{\mathbf{1}}, \widehat{\mathbb{X}}_{t_0,s}^{\alpha^*}\rangle^2\,ds + \int_{t_0}^{t\wedge\tau^{t_0}} \langle \boldsymbol{\psi}_s^{\widehat{\mathbb{X}}_{t_0}^{\alpha^*}}|_{\mathbf{1}}, \widehat{\mathbb{X}}_{t_0,s}^{\alpha^*}\rangle\,dW_s^{\mathbb{Q}^*}.\end{aligned}$$

Recalling that $\psi^*_T = \langle \boldsymbol{p}, \widehat{\mathbb{X}}^{\alpha^*}_T \rangle$ allows us to identify the optimal control in the representation (4.7):

$$\alpha^*_t = \langle \boldsymbol{\psi}_t^{\widehat{\mathbb{X}}^{\alpha^*}_{t_0}} |_{\mathbf{1}}, \widehat{\mathbb{X}}^{\alpha^*}_{t_0,t} \rangle, \quad t \in [\![t_0, \tau^{t_0}]\!]. \tag{4.15}$$

Note that the bound (3.2) implies that $t \mapsto \langle \boldsymbol{\psi}_t^{\widehat{\mathbb{X}}^{\alpha^*}_{t_0}}, \widehat{\mathbb{X}}^{\alpha^*}_{t_0,t} \rangle$ and $t \mapsto \langle \boldsymbol{\psi}_t^{\widehat{\mathbb{X}}^{\alpha^*}_{t_0}} |_{\mathbf{1}}, \widehat{\mathbb{X}}^{\alpha^*}_{t_0,t} \rangle$ are continuous Itô processes and hence admit a continuous modification on $[\![t_0, \tau^{t_0}]\!]$, so that the equalities (4.14) and (4.15) hold for all $t \in [\![t_0, \tau^{t_0}]\!]$ a.s. In particular, positivity of τ^{t_0} implies that the processes ψ^* and α^* admit continuous modifications on $[0, T]$. Thus, (4.3) follows from taking $t \downarrow t_0$ in (4.14) and (4.15) and the fact that t_0 was chosen arbitrary.

The weak existence of the solution to (4.4) follows immideately from Lemma 4.6.

□

5 Numerical implementation

Discuss numerical aspects:

5.1 Scheme for the Riccati equation

5.2 Monitoring and recentering the optimal control

This can actually be made more efficient by continuously keeping track of the condition from Theorem 3.1. For $s \le t \le T$, as long as $\|\boldsymbol{u}_t^{\widehat{\mathbb{W}}_{0,s}}\|_{\widehat{\mathbb{W}}_{s,t}} < 2|\langle \boldsymbol{u}_t^{\widehat{\mathbb{W}}_{0,s}}, \varnothing \rangle|$, we will have

$$\mathbb{E}[\exp(\langle \boldsymbol{p}, \widehat{\mathbb{W}}_{0,T} \rangle) \,|\, \mathcal{F}_t] = \langle \boldsymbol{u}_t^{\widehat{\mathbb{W}}_{0,s}}, \widehat{\mathbb{W}}_{s,t} \rangle = \exp(\langle \boldsymbol{\psi}_t^{\widehat{\mathbb{W}}_{0,s}}, \widehat{\mathbb{W}}_{s,t} \rangle),$$

so that the solution $\boldsymbol{\psi}_\cdot^{\widehat{\mathbb{W}}_{0,s}}$ need not be recomputed until the convergence radius is reached by $\widehat{\mathbb{W}}_{s,t}$.

Remark 5.1. The convergence of $\langle \boldsymbol{\psi}_t, \widehat{\mathbb{W}}_{0,t} \rangle$ is ensured for $t < \tau$, where

$$\tau := \inf\{t\colon \|\exp^{\sqcup\!\sqcup}(\boldsymbol{\psi}_t)\|_{\widehat{\mathbb{W}}_t} \ge 2\exp(\boldsymbol{\psi}_t^{\varnothing})\} = \inf\{t\colon \|\exp^{\sqcup\!\sqcup}(\overline{\boldsymbol{\psi}_t})\|_{\widehat{\mathbb{W}}_t} \ge 2\}.$$

In practice, one cannot numerically evaluate the seminorm $\|\exp^{\sqcup\!\sqcup}(\overline{\boldsymbol{\psi}_t})\|_{\widehat{\mathbb{W}}_t}$ precisely enough, so that a weaker (since $\|\exp^{\sqcup\!\sqcup}(\overline{\boldsymbol{\psi}_t})\|_{\widehat{\mathbb{W}}_t} \le \exp(\|\overline{\boldsymbol{\psi}_t}\|_{\widehat{\mathbb{W}}_t})$), but numerically tractable quantity to track would be

$$\tau^{\boldsymbol{\psi}} := \inf\{t\colon \exp(\|\overline{\boldsymbol{\psi}_t}\|_{\widehat{\mathbb{W}}_t}) \ge 2\} = \inf\{t\colon \|\overline{\boldsymbol{\psi}_t}\|_{\widehat{\mathbb{W}}_t} \ge \log 2\}.$$

■

Remark 5.2. (Explain how to implement the strategy in practice.) If one does not want to monitor continuously the stopping time, one can simple solve a new Riccati equation at each step and set $\alpha^*_{t_i} = \langle \boldsymbol{\psi}_{t_i}^{\widehat{\mathbb{X}}_{t_i}} |_{\mathbf{1}}, \varnothing \rangle$. ■

6 Applications

6.1 Signature tracking

We consider the following control problem:

$$\sup_{\alpha \in \mathcal{A}} \widehat{\mathbb{E}} \left[-\frac{1}{2} \int_0^T |\alpha_t|^2 \, dt - \int_0^T \sum_{k=1}^N \beta_k \|\widehat{\mathbb{X}}^k_t - \widehat{\mathbb{Y}}^k_t\|^2 \, dt \right],$$

for some deterministic process $\widehat{\mathbb{Y}}^{\le N}$ in $T^N(\mathbb{R}^2)$ and a nonnegative vector of coefficients $\beta = (\beta_1, \dots, \beta_N) \in \mathbb{R}^N_+$. Check that this functional verify our conditions. Example: power-tracking $\int_0^T (X^p_s ds - \mu)^2$, with $p \ge 1$. ($p = 1$ std LQ), $p = 2$ (solve numerically?)

We can also consider $\int_0^T (\langle \boldsymbol{\ell}, \widehat{\mathbb{X}}_t \rangle - \mu(t))^2 dt$ with $\boldsymbol{\ell} \in T(\mathbb{R})$ with $\deg \boldsymbol{\ell} = N$ and $\boldsymbol{\ell}^{\mathbf{1}^{\otimes N}} \neq 0$, which forces some path-dependent process of the controlled variable to follow μ.

Signature tracking wiht higher order terms numerics?

6.2 Signature lifts of Volterra control problems

We show how our framework nests certain non-Markovian Volterra control problems. Let Y^α be a controlled Volterra process defined by

$$Y_t^\alpha = y_0 + \int_0^t K(t,s)\big(\alpha_s\, ds + dW_s\big) = y_0 + \int_0^t K(t,s)\, dX_s^\alpha,$$

where X^α is given by (1.1), $y_0 \in \mathbb{R}$ and $K : [0,T]^2 \to \mathbb{R}$ is a measurable kernel satisfying $\sup_{t\le T}\int_0^T K(t,s)^2 ds < \infty$. We consider a cost to be minimized given by

$$J_{\mathrm{Vol}}(\alpha) = \mathbb{E}\left[\int_0^T \left(\frac{\alpha_s^2}{2} + \nu(Y_s^\alpha)^{2m} + \eta(X_s^\alpha)^{2n}\right) ds\right], \tag{6.1}$$

for some $\eta, \nu \ge 0$ and $n, m \in \mathbb{N}$. For $m, n \in \{0,1\}$, the problem reduces to a linear–quadratic Volterra control problem, which admits a semi-explicit solution and provides a natural benchmark, see Proposition 6.2 below. Variants of such problems in the linear-quadratic setting have been studied in various applications, notably in optimal execution with transient market impact; see (Abi Jaber, Neuman, and Voß, 2023). For $m \ge 2$, no general explicit solution is known, and classical dynamic programming approaches fail due to the lack of a finite-dimensional Markovian state.

We now show how this class of Volterra control problems can be embedded into our framework by writing J_{Vol} in the form TO REF. The key observation is that, for a broad class of kernels, the Volterra process Y^α admits a representation as a linear functional of the time-augmented signature of the driving process X^α, namely

$$Y_t^\alpha = \langle \boldsymbol{k}_t, \widehat{\mathbb{X}}_t^\alpha \rangle, \tag{6.2}$$

for some coefficient sequence $\boldsymbol{k}_t$, which may be either time-dependent or time-independent and may contain finitely or infinitely many nonzero terms. Such representations have recently been derived in Abi Jaber, Gérard, and Huang (2024a) for broad classes of kernels.

The following proposition provides two important examples.

Proposition 6.1. *For kernels of the form $K(t,s) = \sum_{n=0}^N k_i(t)\frac{s^i}{i!}$, for some coefficients $(k_i(t))_{i\ge 0}$, the representation (6.2) holds with time-dependent coefficients*

$$\boldsymbol{k}_t = y_0 \varnothing + \sum_{i=0}^N k_i(t)\mathbf{0}^{\mathbf{i}}\mathbf{1}.$$

For kernels of the form $K(t,s) = \sum_{i=1}^N c_i e^{-\lambda_i(t-s)}$, for some $c_i, \lambda_i \in \mathbb{R}$. Then the representation (6.2) holds with time-independent coefficients

$$\boldsymbol{k} = \sum_{i=1}^N c_i\big(y_0\varnothing + \mathbf{1}\big)e^{\sqcup\!\sqcup(-\lambda_i \mathbf{1})}.$$

Proof. We only sketch the proof of the first case. Note that for $i \ge 0$,

$$\frac{s^n}{i!} = \int_{0<s_1<\ldots<s_i} ds_1 \ldots ds_i = \langle \mathbf{0}^i, \widehat{\mathbb{X}}_s^\alpha \rangle,$$

so that $\int_0^t \frac{s^i}{i!} dX_s^\alpha = \int_0^t \langle \mathbf{0}^i, \widehat{\mathbb{X}}_s^\alpha\rangle dX_s^\alpha = \langle \mathbf{0}^i\mathbf{1}, \widehat{\mathbb{X}}_t^\alpha\rangle$. Substituting the power series expansion of K into the definition of Y^α and exchanging summation and integration yields

$$Y_t^\alpha = y_0 + \sum_{i=0}^N k_i(t)\int_0^t \frac{s^i}{i!} dX_s^\alpha = y_0 + \sum_{i=0}^N k_i(t)\langle \mathbf{0}^i\mathbf{1}, \widehat{\mathbb{X}}_t^\alpha\rangle$$

which proves the first point. The second case follows directly from the signature representation of Ornstein Uhlenbeck-type processes as done in (Abi Jaber et al., 2024a, Example 4.3). □

Remark 6.1. The two examples in Proposition 6.1 cover broad classes of kernels and also provide natural approximation schemes. For instance, fractional kernels of the form $K(t,s) = (c+t-s)^{\beta-1}\mathbf{1}_{s<t}$, with $c \geq 0$ and $\beta > 0$, admit an infinite power series expansion of the form

$$K(t,s) = \sum_{i=0}^{\infty} (c+t)^{\beta-1-i}(-1)^i \frac{s^i}{i!},$$

and truncating at finite order yields a natural approximation. Likewise, finite sums of exponentials form a flexible class that can approximate a wide range of kernels. We refer to Abi Jaber et al. (2024a) for further examples and discussion.

As a consequence, by Proposition 6.1 and the shuffle property, polynomial costs in Y^α can be rewritten as polynomial functionals of the signature of X^α, so that the Volterra control problem (6.1) falls within our framework:

$$J_{\mathrm{Vol}}(\alpha) = \mathbb{E}\left[\int_0^T \left(\frac{\alpha_s^2}{2} + \langle \boldsymbol{q}_s, \widehat{\mathbb{X}}_s^\alpha \rangle\right) ds\right],$$

with

$$\boldsymbol{q}_s = (2m)!\eta \boldsymbol{k}_s^{2m} + (2n)!\nu(\mathbf{1}\mathbf{1})^{2n},$$

where $\boldsymbol{k}_s$ is given as in Proposition 6.1. In particular, whenever the orders m, n and coefficients are chosen so that our main Theorem 4.3 applies, this yields a tractable representation and implementation of the optimal control for a nontrivial class of genuinely nonlinear and non-Markovian Volterra control problems. While our current theory is formulated under specific structural assumptions within the class $\mathcal{B}$, the above representation remains valid more generally and suggests that the scope of the method extends beyond the present setting.

DS: $(2n)!\nu(\mathbf{1})^{\sqcup}$ Or $C\nu(\mathbf{11})^{\sqcup\sqcup_n}$

eaj: verifier et expliciter les hypotheses?

Peut être pour plus tard si on faire du numerique

Solution for $m = 1$. To state the solution, we introduce some notations. For a kernel K we denote by $\boldsymbol{K}$, the integral operator from L^2 into itself induced by the kernel K, i.e. $(\boldsymbol{K}f)(t) = \int_0^T K(t,s)f(s)ds$, and by $\boldsymbol{K}^*$ the adjoint of K for the inner product $\langle f, g\rangle_{L^2} := \int_0^T f(s)g(s)ds$ for square integrable functions f, g, i.e. $\boldsymbol{K}^*$ is induced by the kernel $K^*(s,u) = K(u,s)$. We denote by id the identity operator, that is $(\mathrm{id}f)(t) = f(t)$. Finally, det denotes the Fredholm determinant of a trace class operator.

We define

$$g_t(s) = 1_{s<t}\left(y_0 + \int_0^t K(s,u)dW_u\right),$$

and $\boldsymbol{\Sigma_t}$ the integral operator induced by the covariance kernel

$$\Sigma_t(s,r) = 1_{t\leq s,r}\int_t^{s\wedge r} K(s,u)K(r,u)du.$$

The following proposition provides a semi-explicit solution in terms of Fredholm operators and resolvents, for the case $m = 1$. The solution can be derived using variational approach, leading to linear stochastic Fredholm equation, see for instance REF. For completeness, we provide a sketch of proof using the Boué-Dupuis formula.

Proposition 6.2. *For $m = 1$,*

$$\inf_{\alpha\in\mathcal{A}^2} \mathbb{E}\left[\int_0^T \left(\frac{1}{2}\alpha_s^2 + (Y_s^\alpha)^2\right) ds\right] = \frac{1}{2}\log\det(\mathrm{id} + 2\boldsymbol{\Sigma}_0) + y_0^2\langle 1_t, (\mathrm{id} + 2\boldsymbol{\Sigma}_0)^{-1}1_t\rangle_{L^2}, \tag{6.3}$$

where $1_t : s \mapsto 1_{t\leq s}$, and the unique optimal control α^* is given by

$$\alpha_t^* = -2\left(\boldsymbol{K}^*(\mathrm{id} + 2\boldsymbol{\Sigma}_t)^{-1}g_t\right)(t), \quad t \in [0,T]. \tag{6.4}$$

to double check wether g or g^α

Sketch of proof. An application of the Boué-Dupuis to check formula yields

$$\inf_{\alpha\in\mathcal{A}^2_{t,T}} \mathbb{E}\left[\int_t^T \left(\frac{1}{2}\alpha_s^2 + (Y_s^\alpha)^2\right) ds \;\middle|\; \mathcal{F}_t\right] = -\log \mathbb{E}\left[\exp\left(-\int_t^T Y_s^2 ds\right) \;\middle|\; \mathcal{F}_t\right] \tag{6.5}$$

where Y is the Gaussian process

$$Y_t = y_0 + \int_0^t K(t,s) dW_s.$$

Note that $g_t(s) = \mathbb{E}[Y_s|\mathcal{F}_t]$ and $\Sigma_t(s,u) = \mathbb{E}[(Y_s - g_t(s))(Y_u - g_t(u))|\mathcal{F}_t]$ are the conditional mean and covariance of Y. Now, the conditional Laplace transform of $\int_t^T Y_s^2 ds$ can be readily computed:

$$\mathbb{E}\left[\exp\left(-\int_t^T Y_s^2 ds\right) \;\middle|\; \mathcal{F}_t\right] = \exp\left(-\frac{1}{2}\log\det(\mathrm{id} + 2\boldsymbol{\Sigma}_t) - \langle g_t, (\mathrm{id} + 2\boldsymbol{\Sigma}_t)^{-1} g_t\rangle_{L^2}\right)$$

see for instance (Abi Jaber, 2022, Theorem 2.3). Combining with (6.5) and evaluating at $t = 0$, yields (6.3). Differentiation yields (6.4). By strict convexity of the control problem, α^* is the unique optimal control. □

DS: Is it true for GPs (other than the BM?)

eaj :we can derive the solution differently, but i think yes, at least formally Y is a linear functional of the Brownian motion W.

6.3 Importance of recentering

References

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